

Appendix. Published benchmarks for material demand responses

Table A1. Published benchmarks for material demand responses.

Study	Outcome	Income or		Interpretation
		Price response	activity response	
This study	Korean apparent steel consumption	−0.272	+0.799	Converged GMM response
Fernandez (2018a)	World steel consumption	−0.086	+1.156	IV system in log levels
Tcha and Takashina (2002)	World steel consumption	−0.012	+1.030	Contemporaneous differences with a constant and Divisia aggregates
Crompton (2015)	Steel consumption in 26 OECD countries	Not estimated	0.01 to 4.05	Country income elasticities from a static panel
This study	Korean apparent cement consumption	−0.548 to −0.367	Not part of this set	Conditional point response set under the signed bound of 0.05
Seo (2021)	Korean domestic cement shipments	0.842 total cement and 0.977 Portland cement	Not estimated	Monthly IV-ECM; absolute elasticities
Stuermer (2017)	Five base metals in 15 countries	−0.691 to −0.035	+0.428 to +1.495	Long-run PMG; manufacturing output

Notes. Estimates differ in sample, outcome, frequency, and method and do not form a pooled range. Tcha and Takashina use Divisia relative metal prices and an aggregate metal quantity index rather than GDP. Crompton reports variation across countries. Stuermer reports ranges across aluminum, copper, lead, tin, and zinc, with activity measured by real manufacturing value added per capita. Seo reports absolute magnitudes, not positive demand slopes. The present cement set assumes the signed direct-effect bound $c = 0.05$ and is not a confidence interval. Uncertainty for this study is reported in Section 5.